Philosophy Engineering: Executable, Falsifiable, Auditable Machine Ethics

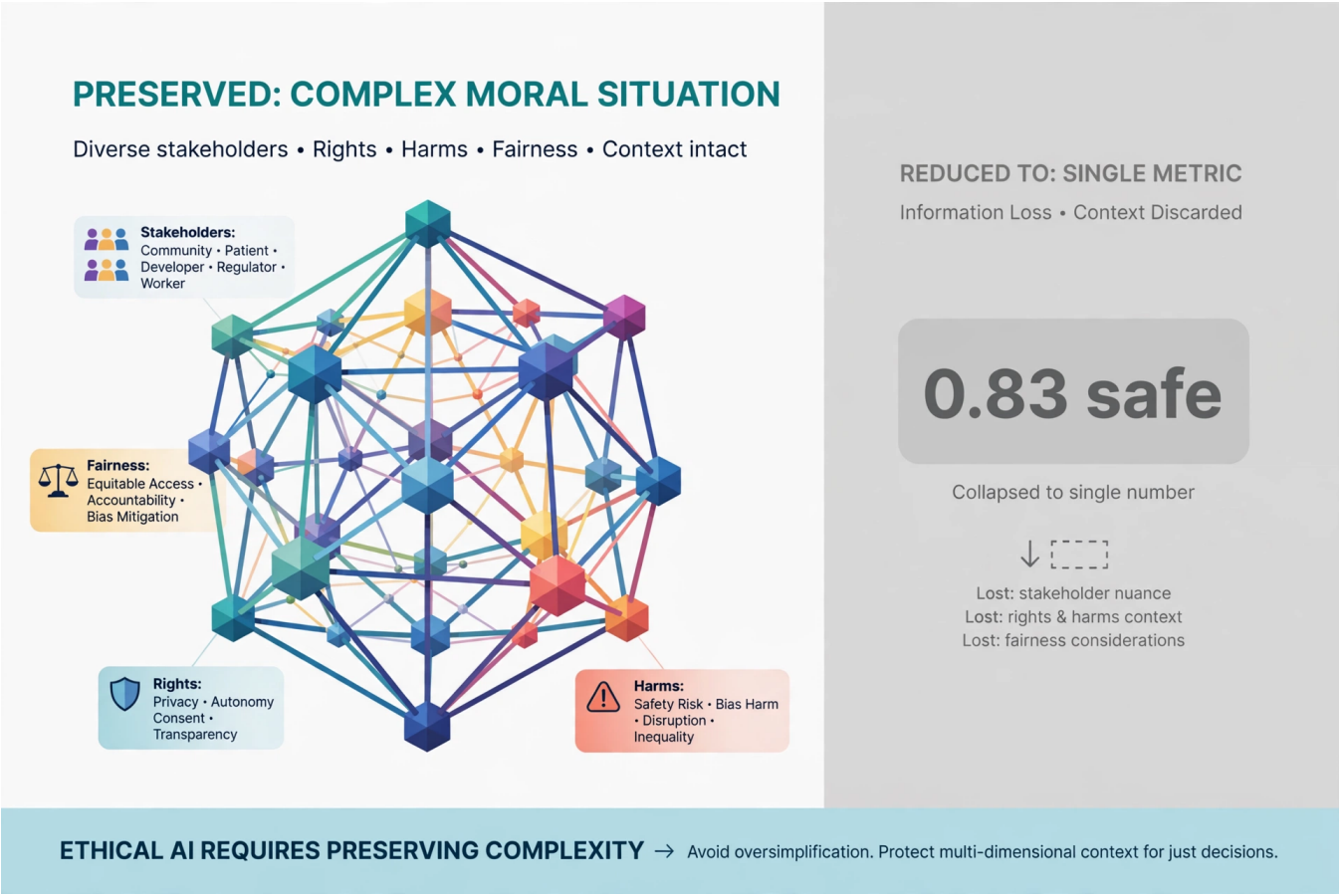
ErisML + DEME — A Structure-Preserving, Framework-Pluralist Pipeline — Layered with RLHF, Not Replacing It

30-Second Pitch
Deployed AI collapses moral reasoning into a single number — '0.83 safe' — that discards structure and silently pre-commits to one ethical framework while appearing neutral. We preserve the structure: compile situations into a typed, hash-chained MoralGraph, evaluate with a 9D moral tensor, run plural frameworks with honest refusal to aggregate, and route disagreement to human review. Audit trail is a property of computation.

2-Minute Pitch
RLHF / Constitutional AI shapes model *dispositions* — helpful, harmless. That's necessary but insufficient. ErisML + DEME shapes *accounting*: L0 Base LLM, L1 Dispositional shaping, L2 Structure-preserving compiler (6 node kinds, 11 edge kinds, SHA-256 canonical), L3 DEME graded valuation + per-stakeholder tensor + tier-0 Geneva hard veto, L4 Framework pluralism (consequentialist, Kantian SMT-checked, virtue, care) with worst-off/escalate policy, L5 Human review, L6 Hash-chained DecisionProof ledger. Scalar collapse theorem: no continuous injection [0,1]^d->R for d>=2. Scalar is last step, not substrate.

The Problem: Scalar Collapse Loses What Matters

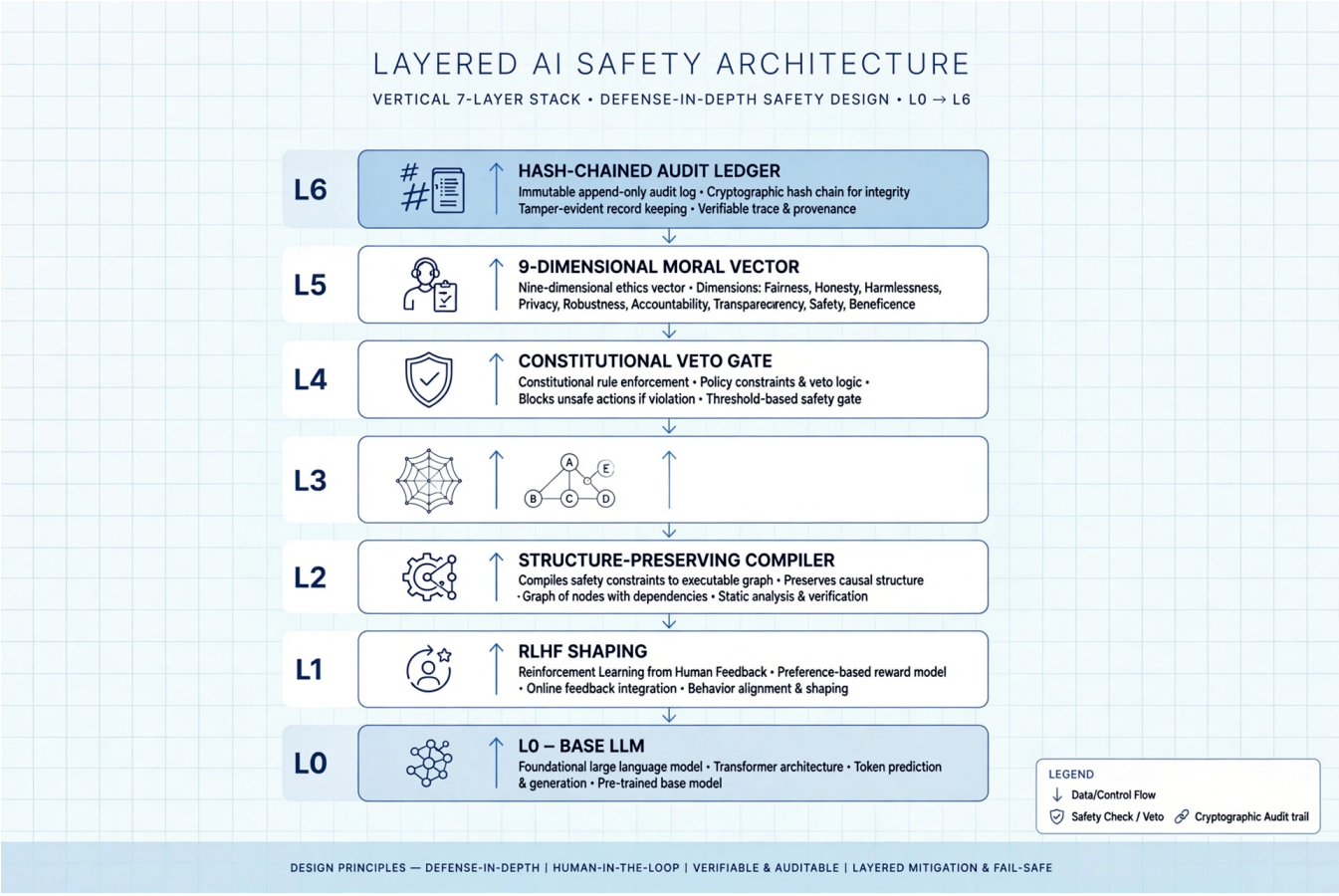
Moral situations are plural: who is harmed, under which maxim, with what consent, in tension with whose rights. Reducing to reward score or guardrail pass/fail is moral monism — Williams' critique of aggregation, Ross's prima facie duties — and is information-theoretic lossy. No lossless scalar option. Choosing to collapse is a normative choice that must be auditable.



Left: Preserved complex moral situation — stakeholders, rights, harms, fairness intact as graph. Right: Reduced to 0.83 safe — stakeholder nuance, rights & harms context, fairness considerations lost. Ethical AI requires preserving complexity.

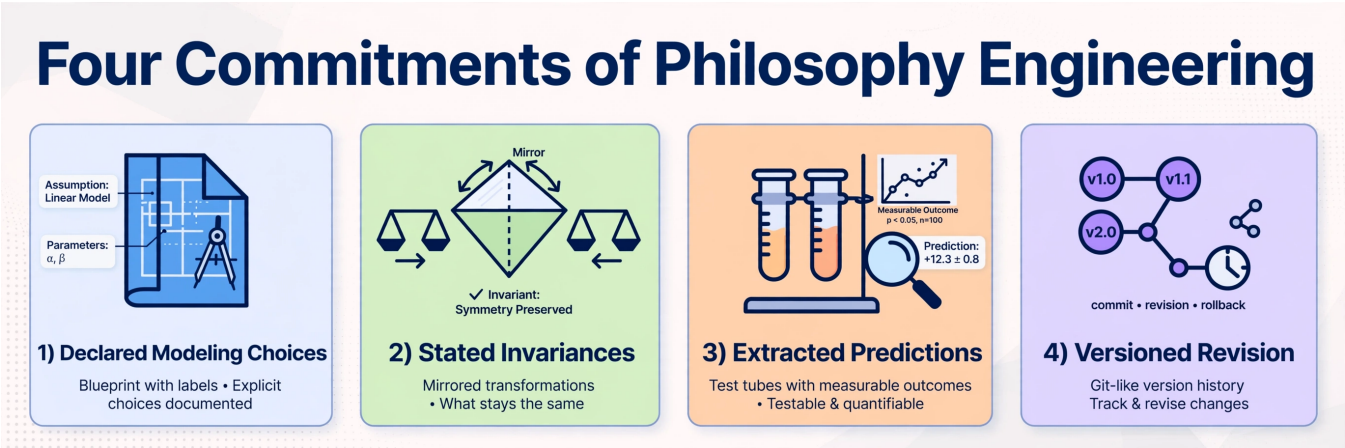
The Solution: Layered Safety — Orthogonal, Not Replacement

RLHF is L1. ErisML/DEME is L2-L6 audit substrate. Layer them defense-in-depth. Fast reflex veto can run pre-filter; full graph + SMT only on flagged high-stakes cases. 91% escalation on all-dilemmas distribution is triage rate, not deployment failure — benign cases where frameworks agree to permit dominate real streams.



L0 Base LLM (Transformer) | L1 RLHF Shaping (preference reward, behavior alignment) | L2 Structure-Preserving Compiler (compiles constraints to executable graph, static verification) | L3 Moral Graph IR | L4 Constitutional Veto Gate (policy constraints, blocks unsafe if violation) | L5 9-Dimensional Moral Vector | L6 Hash-Chained Audit Ledger (immutable, tamper-evident). Data/control flow ↓, Safety check/veto, Cryptographic audit trail.

Methodology: Four Commitments of Philosophy Engineering



1. Declared Modeling Choices
Every bridge moral claim -> computational object carries epistemic status: definition, assumption, conditional theorem, empirical result. Proven vs posited not confused.

2. Stated Invariances
Transformations under which evaluation must be stable — meaning-preserving re-description — written down and tested, not assumed. 13.7% euphemistic shift, 16.1% dramatizing shift surfaced as invariance violations.

3. Extracted Predictions
Reliability ICC(2,k)=0.97 α=0.84 inter-model agreement (6 LLMs, 31 scenarios, n=217), test-retest r=0.87-0.98. Robustness residual drift θd 0.41->0.22 reframing, 0.48->0.30 back-translation. Formal gate SMT-checked conditional on declared formalization.

4. Versioned Revision
When prediction fails, model amended, change recorded. Example: moral reference = Fréchet mean on curved manifold, schism = bifurcation beyond Karcher-Afsari radius — falsified on GlobalOpinionQA & Moral Machine, diagnostic kept, conjecture retired. Honest negatives at equal prominence.

ErisML + DEME in One Line

Structure-preserving compiler parses NL scenarios into typed MoralGraph DAG (stakeholder, act, maxim, commitment, fact, norm) canonically hashed. Plurality of framework projections (consequentialist, Kantian with Z3 universalizability gate, virtue, care) refusing to aggregate when disagree — disagreement is first-class caller-resolved metaethical event. DEME maps to canonical 9D moral vector [physical harm, rights, fairness, autonomy, privacy, societal/env, virtue/care, legitimacy/trust, epistemic quality] extended to per-stakeholder tensor, tier-0 constitutional modules hold hard veto (Geneva). Out-of-band I-EIP monitor only sanctioned failure output: requires human review. Hash-chained DecisionProof links verdict to graph identity and module weights. Open, installable.

Why Now
LLMs made normative reasoning deployable at scale and consequence — raises stakes of methodology. Governance specs (model cards, NIST AI RMF, IEEE 7000, EU AI Act) say what to disclose, not mechanism that produces inspectable reasoning.

Workshop: Philosophy Engineering @ AIES 2027
We propose full-day workshop bridging machine ethics, alignment, formal ethics, computational social choice, governance. Format: keynotes, contributed talks, hands-on building & falsifying normative model, lightning + critiques, panel 'Can ethics be engineered without being reduced?', community roadmap. Non-archival. Seeking 3-5 co-organizers multi-institution. Contact: agi.hpc@gmail.com

Join / Install
Open artifacts, reproducible notebooks, drift-guarded CI. Lead: Andrew H. Bond, SJSU. MoralGraph IR, DEME engine, audit ledger. Falsifiable completeness claim: exhibit genuine moral consideration independent of all nine and not combination, basis must extend (Purity extension absorbed versioned, not breaking).